Chapter 4. Learning to pick the best policy: Policy gradient methods

Figure 4.1. A Q-network takes a state and returns Q values (action values) for each action. We can use those action values to decide which actions to take.

Figure 4.2. A policy network is a function that takes a state and returns a probability distribution over the possible actions.

Figure 4.3. A stochastic policy function. A policy function accepts a state and returns a probability distribution over actions. It is stochastic because it returns a probability distribution over actions rather than returning a deterministic, single action.

Figure 4.4. A deterministic policy function, often represented by the Greek character pi, takes a state and returns a specific action to take, unlike a stochastic policy, which returns a probability distribution over actions.

Figure 4.5. The general overview of policy gradients for an environment with four possible discrete actions. First we input the state to the policy network, which produces a probability distribution over the actions, and then we sample from this distribution to take an action, which produces a new state.

Figure 4.6. Once an action is sampled from the policy network's probability distribution, it produces a new state and reward. The reward signal is used to reinforce the action that was taken, that is, it increases the probability of that action given the state if the reward is positive, or it decreases the probability if the reward is negative. Notice that we only received information about action 3 (element 4), but since the probabilities must sum to 1, we have to lower the probabilities of the other actions.

Figure 4.7. A string diagram for training a policy network for Gridworld. The policy network is a neural network parameterized by θ (the weights) that accepts a 64-dimensional vector for an input state. It produces a discrete 4-dimensional probability distribution over the actions. The sample action box samples an action from the distribution and produces an integer as the action, which is given to the environment (to produce a new state and reward) and to the loss